MULTIPLE INPUT LINEAR REGRESSION

[1]:
```
import pandas as pd
```

[2]:
```
dataset = pd.read_csv('Startups_dataset.csv')
#dataset
```

[3]:
```
dataset = dataset.drop(columns=["Unnamed: 0","State"])
```

[4]:
```
#dataset
```

[5]:
```
dataset.isnull().sum()
```

[5]:
```
R&D Spend          0
Administration     0
Marketing Spend    0
Profit             0
dtype: int64
```

[6]:
```
x = dataset[["R&D Spend", "Administration", "Marketing Spend"]]
y = dataset[["Profit"]]
```

[7]:
```
#x
```

[8]:
```
import matplotlib.pyplot as plt
plt.scatter(x["R&D Spend"],y)
plt.scatter(x["Administration"], y)
plt.scatter(x["Marketing Spend"], y)
plt.show()
```

[9]:
```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random_state=39)
```

[10]:
```
# y_test
```

[11]:
```
from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(x_train, y_train)
```

[11]:

LinearRegression

▸ Parameters

▸ Fitted attributes

[12]:
```
y_pred = regressor.predict(x_test)
# y_pred
```

[13]:
```
from sklearn.metrics import r2_score
result_accuracy = r2_score(y_test, y_pred)
result_accuracy
```

[13]:
```
0.9704550565307896
```

[14]:
```
profit = regressor.predict([[123456, 2345, 1234987]])
profit
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does
was fitted with feature names
  warnings.warn(
```

[14]:
```
array([[180182.02085944]])
```

[15]:
```
import pickle
filename = "startup.sav"
pickle.dump(regressor, open(filename, "wb"))
```

.py File

```python
import streamlit as st
import pickle
import numpy as np
import pandas as pd

model = pickle.load(open("startup.sav","rb"))
st.set_page_config()

st.title("Startup Profit detection app")
st.write("Predict an startup profit based on their expense")
st.divider()
rd_spend = st.number_input(
    "Enter R&D Spend",
    min_value = 0.0,
    max_value = 999999.0,
    value = 1.0,
    step = 0.01
)
admin = st.number_input(
    "Enter the Administration",
    min_value = 0.0,
    max_value = 999999.0,
    value = 1.0,
    step = 0.01
)
mark = st.number_input(
    "Enter the Marketing Spend",
    min_value = 0.0,
    max_value = 999999.0,
    value = 1.0,
    step = 0.01
)
```

```python
if st.button("Predict Profit"):
    input_data = pd.DataFrame([{
        'R&D Spend': rd_spend,
        'Administration': admin,
        'Marketing Spend': mark
    }])
    prediction = model.predict(input_data)
    predict_profit = prediction[0][0]
    st.success(
        f"Predicted Profit: ₹{float(predict_profit):,.2f}"
    )
```